3D PRINTING &

ADDITIVE MANUFACTURING TECHNOLOGY

Quadruped Robot Development Report

Report of Quadruped Robot Development

Department of Electronics and Communication Engineering

Cooch Behar Government Engineering College

2026

Certificate of Approval

It is hereby approved that the project report entitled **Report of Quadruped Robot Development** submitted by:

1. SARUPYA GUHA (Roll No.: 34900323046 / Reg. ID: FS08A0198)
2. SOHAN GHOSH (Roll No.: 34900323050 / Reg. ID: FS08A0202)
3. SAYAN SARKAR (Roll No.: 34900323048 / Reg. ID: FS08A0157)
4. DEBANJAN CHAKRABORTY (Roll No.: 34900323015 / Reg. ID: FS08A0154)
5. JOYDIP DAS (Roll No.: 34900323020 / Reg. ID: FS08A0220)
6. AUSHI SARKAR (Roll No.: 34900323011 / Reg. ID: FS08A0288)
7. ANUPAM DUTTA (Roll No.: 34900323003 / Reg. ID: FS08A0206)
8. BIKKRAM DAS (Roll No.: 34900323012 / Reg. ID: FS08A0528)
9. BIRAJ SARKAR (Roll No.: 34900323013 / Reg. ID: FS08A0239)
10. SOUMYAJIT PANDIT (Roll No.: 34900323052 / Reg. ID: FS08A0189)

As part of their certification-based industrial training in **3D PRINTING & ADDITIVE MANUFACTURING TECHNOLOGY**, conducted under the supervision of **WTL & MeitY, Cooch Behar Government Engineering College**, Cooch Behar, West Bengal.

Dr. Palash Das	Mrs. Susmita Banik Barik
(HOD, Dept. of ECE, CGEC)	(Project Engineer, WTL, MeitY)

REPORT FOR QUADRUPED ROBOT DEVELOPMENT

Certification-Based Industrial Training

WTL & MeitY Project, CGEC

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SOUMYAJIT PANDIT (Roll No.: 34900323052 / Reg. ID: FS08A0189)

Group-E

Under the guidance of:

Dr. PALASH DAS
(HOD of ECE Dept., CGEC)

Mrs. SUSMITA BANIK BARIK
(Project Eng., WTL & MeitY Project)

Mr. ARIJIT BHUNIA
(Project Co-ordinator, WTL & MeitY Project)

Mr. BHASKAR MONDAL
(Faculty, WTL & MeitY Project)

Cooch Behar Government Engineering College

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1. 3D Printing Process

1.1. What is Additive Manufacturing?

Additive Manufacturing (AM) is a process of building a three-dimensional object by depositing material layer-by-layer directly from a digital model, as opposed to subtractive manufacturing which removes material from a solid block. AM enables fabrication of complex geometries that are difficult or impossible to produce by conventional methods, with minimal material waste.

1.1.1. Types of Additive Manufacturing

- **Vat Photopolymerization (SLA / DLP)** — A liquid photopolymer resin is selectively cured by a UV light source (laser or digital projector) layer by layer. The build platform moves upward (or downward in bottom-up systems) after each layer. Produces very high resolution parts but resins are brittle and sensitive to UV degradation. Used in jewellery, dental, and prototyping.

- **Powder Bed Fusion (SLS / SLM / DMLS)** — A thin layer of powder (polymer, metal, or ceramic) is spread across a build platform and a laser or electron beam fuses the powder at selected points. Unfused powder acts as support. SLS uses nylon/PA12; SLM/DMLS use metal alloys such as titanium and stainless steel. Used extensively in aerospace, medical implants, and high-performance engineering parts.

- **Fused Deposition Modelling (FDM / FFF)** — A thermoplastic filament is fed through a heated nozzle and deposited in a predetermined path on the build platform, bonding to the previous layer as it cools. This is the most widely used AM process for education and prototyping due to its low cost, ease of use, and wide range of compatible materials. **All parts in this project were printed using FDM.**

- **Material Jetting** — Droplets of photopolymer are jetted onto the build surface and cured immediately by UV. Allows multi-material and full-colour prints. High accuracy but expensive.

- **Binder Jetting** — A liquid binder is selectively deposited onto a powder bed to join particles. No heat is applied during printing; a post-process sintering step densifies the part. Used for sand casting moulds and metal parts.

- **Directed Energy Deposition (DED)** — A focused energy source (laser or electron beam) melts material (wire or powder) as it is deposited. Mainly used for repairing or adding features to existing metal components.

1.2. Subtractive Manufacturing — A Brief Overview

Unlike additive manufacturing, subtractive manufacturing starts with a solid block (workpiece) and removes material using cutting tools to achieve the desired shape. Key processes include:

- **CNC Milling** — A rotating multi-point cutter removes material along multiple axes. Produces flat surfaces, slots, and complex 3D contours in metal, wood, and plastic.

- **CNC Turning (Lathe)** — The workpiece rotates against a fixed cutting tool. Used for cylindrical parts such as shafts, bushings, and threaded components.

- **Drilling** — A rotating drill bit creates holes of defined diameter and depth.

- **Grinding** — An abrasive wheel removes small amounts of material for high surface finish and tight tolerances.

- **EDM (Electrical Discharge Machining)** — Material is removed by controlled electrical sparks between electrode and workpiece. Used for hardened metals and complex cavities.

Compared to subtractive methods, FDM 3D printing requires no fixturing, produces no chips, and can create internal cavities freely — making it ideal for rapid prototyping of robot chassis components.

1.3. 3D File Formats

A 3D model must be saved in a compatible file format before it can be sliced and printed. Common formats include:

- **STL (Standard Triangle Language / Standard Tessellation Language)** — Represents the surface geometry of a 3D object as a collection of linked triangles (tessellation). Each triangle is defined by three vertices and a normal vector. STL files are simple, widely supported, and easily understood by 3D printers. They do not store colour, texture, or material information. **In our training, we exclusively use `.stl` files** exported from FreeCAD for all printed parts.

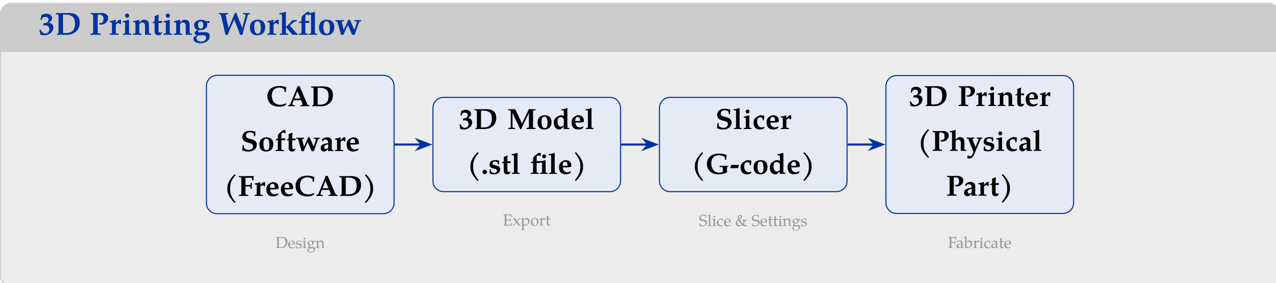
- **OBJ (Wavefront OBJ)** — A text-based format storing vertices, normals, texture coordinates, and polygon faces. Supports colour and texture via an associated `.mtl` material file. More complex than STL but better suited for visual rendering and multi-colour models.

- **VRML / X3D (.wrl / .x3d)** — Virtual Reality Modelling Language. Stores 3D geometry along with colour, texture, and scene information. Used for interactive 3D web content and full-colour printing on specialised printers.

- **3MF (3D Manufacturing Format)** — A modern XML-based format developed by Microsoft and industry partners. Stores geometry, colour, material, and print settings in a single file. Increasingly adopted as a replacement for STL in professional workflows.
- **STEP / IGES** — CAD exchange formats that store precise parametric geometry (B-rep). Used for engineering data exchange between CAD packages, not directly by slicers.

1.4. 3D Printing Workflow

The complete workflow from design to physical part involves the following stages:



1. **CAD Software** — Design the part in a CAD tool such as FreeCAD, SolidWorks, or Fusion 360 using parametric modelling.
2. **3D Model Export** — Export the completed body as an `.stl` file (tessellated surface mesh).
3. **Slicer** — Import the STL into slicer software (Cura, PrusaSlicer, etc.). Set parameters: layer height, infill, walls, supports, temperature. The slicer generates G-code — a sequence of movement and extrusion instructions for the printer.
4. **3D Printer** — Transfer G-code via SD card or USB. The printer executes the G-code, depositing molten filament layer by layer until the part is complete.

1.5. Materials Used in FDM 3D Printing

- **PLA (Polylactic Acid)** — Biodegradable, plant-based polymer. Easy to print, low warp, good stiffness. Brittle under impact. Best for display models and rigid structural parts. Print temp: 190–220°C.
- **PLA+ (Enhanced PLA)** — Improved PLA with toughness additives. Better layer adhesion and impact resistance than standard PLA while retaining ease of printing. **Used for Core Plate, Head Box, and Servo Mounts in this project.**
- **ABS (Acrylonitrile Butadiene Styrene)** — Tough, impact resistant, heat tolerant (up

to ~100°C). Prone to warping; requires heated bed and enclosure. Used in automotive and functional parts. Print temp: 220–250°C.

- **PETG (Polyethylene Terephthalate Glycol)** — Combines PLA's ease of printing with ABS's toughness and chemical resistance. Low warp, food-safe grades available, good layer bonding. Print temp: 230–250°C. **Used for leg components in this project.**
- **TPU (Thermoplastic Polyurethane)** — Flexible, rubber-like filament. Used for gaskets, phone cases, and flexible hinges.
- **PETG / Nylon / PC** — Advanced engineering filaments for high-temperature or high-stress applications.

1.6. Print Parameters Comparison

Parameter	**Low Setting**	**Medium Setting**	**High Setting**
Layer Height	0.05–0.1 mm (fine detail, slow)	0.15–0.2 mm (balanced)	0.25–0.35 mm (draft, fast)
Print Resolution	High surface quality, visible layers minimal	Standard finish, visible layers moderate	Rough finish, visible layers prominent
Object Height	Same height takes 3–4× longer	Moderate print time	Fastest build rate, lower accuracy
Print Time	Very long (hours)	Moderate	Short

In this project we used 0.2 mm layer height as it provides a good balance of surface quality and print time.

1.7. Infill Patterns and Print Paths

The **infill pattern** defines the internal structure of a printed part. The design of the infill pattern determines the strength, flexibility, and material usage. Common patterns include:

- **Grid** — Two sets of perpendicular lines forming a square grid. Good all-round strength, commonly used for structural parts.
- **Lines** — Parallel lines alternating direction each layer. Fast to print, lower strength.
- **Triangles** — Triangular grid providing good shear strength along horizontal planes.
- **Honeycomb / Hexagonal** — Hexagonal cells provide excellent strength-to-weight ratio, similar to natural honeycomb structures.

- **Concentric** — Rings following the outer wall shape. Good for flexible parts and top/bottom surface quality.
- **Gyroid** — Complex three-dimensional wave surface. Excellent isotropic strength in all directions; used for functional parts.

The **print path** (toolpath) defines how the nozzle moves to deposit material:

- **Raster Path** — The nozzle sweeps back and forth in parallel lines (like a raster scan). Simple and fast.
- **Grid Path** — Combines two raster passes at 90° to each other for a crossed grid structure.
- **Contour Path** — The nozzle follows the outline of the part cross-section, then fills inward. Used for perimeter walls to give a clean outer finish.

In this project we used 40% infill with a **grid pattern** for structural parts (Core Plate, Head Box) and **lines** for the slender leg components to reduce weight.

1.8. Slicer Settings Used in This Project

Parameter	**Value**
Layer Height	0.2 mm
Infill Density	40%
Infill Pattern	Grid
Perimeter Walls	4
Nozzle Diameter	0.4 mm
Print Speed	50 mm/s
Support	As required

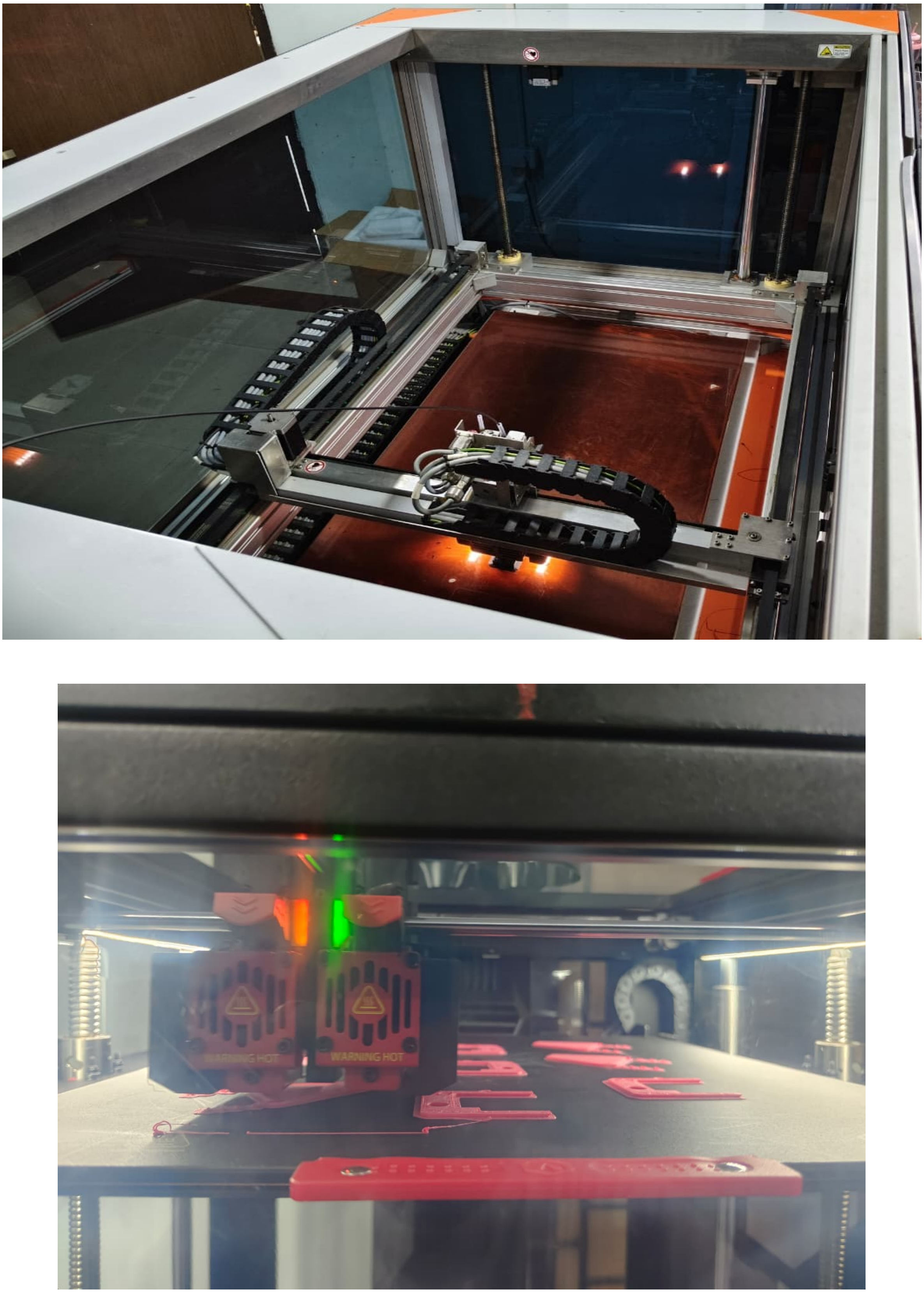


Figure 1: 3D Printer during a print job

2. Objective

The primary objective of this project is to design, fabricate, and program a fully functional **8-DOF Quadruped Robot** using affordable components and open-source tools. The robot uses an **ESP32 microcontroller**, a **PCA9685 16-channel servo driver**, and eight **SG90 micro servo motors**. The chassis is designed in **FreeCAD** and fabricated using **FDM 3D printing** with PLA+ and PETG filament.

The project gives Group-E students hands-on experience across three domains: **electronics** (I2C communication, servo control), **mechanical design** (parametric CAD, dimensional constraints), and **additive manufacturing** (slicing, material selection, print optimisation). The goal is a robot capable of standing stably and executing a walking gait, with a roadmap extending towards sensor integration and autonomous navigation.

3. Introduction to Quadruped Robotics

3.1. What is a Quadruped Robot?

A quadruped robot is a legged mobile robot that moves on four limbs, mimicking the locomotion of four-legged animals such as dogs, horses, and insects. Unlike wheeled robots, quadrupeds can navigate uneven terrains, climb stairs, step over obstacles, and operate in environments where conventional wheeled platforms fail. Each leg typically has two or more joints, providing the degrees of freedom needed for stable gait generation. In this project, each leg has two joints — a **hip joint** and a **knee joint** — giving the robot 8 DOF for locomotion, plus one additional DOF for head rotation.

3.2. Real-World Applications

Quadruped robots have found applications across research, industry, and defence:

- **Search and Rescue:** Navigating rubble where wheels cannot go.
- **Rough Terrain Inspection:** Pipeline inspection and remote surveys.
- **Military and Defence:** Cargo carrying and reconnaissance.
- **Medical and Rehabilitation:** Companion and assistive devices.
- **Research Platform:** Studying locomotion, control, and AI navigation.

Notable examples include Boston Dynamics' Spot, Unitree's Go1, and numerous university-built platforms.

3.3. Why ESP32?

The **ESP32** was chosen as the central controller because it is low-cost, powerful (dual-core 240 MHz), includes Wi-Fi and Bluetooth natively, supports I2C and SPI, and has a well-supported Arduino IDE ecosystem. Its built-in `ESP32Servo` library makes direct servo control straightforward. The wireless capability opens the door to remote teleoperation and OTA firmware updates in future phases.

3.4. Additive Manufacturing Angle

Using 3D printing for the chassis allows rapid iteration on dimensions, customisation of mounting positions, and fabrication of geometrically complex parts that would be expensive to machine. PLA+ offers excellent stiffness for structural plates, while PETG provides the flexibility and impact resistance needed for dynamic leg components subject to repeated bending forces.

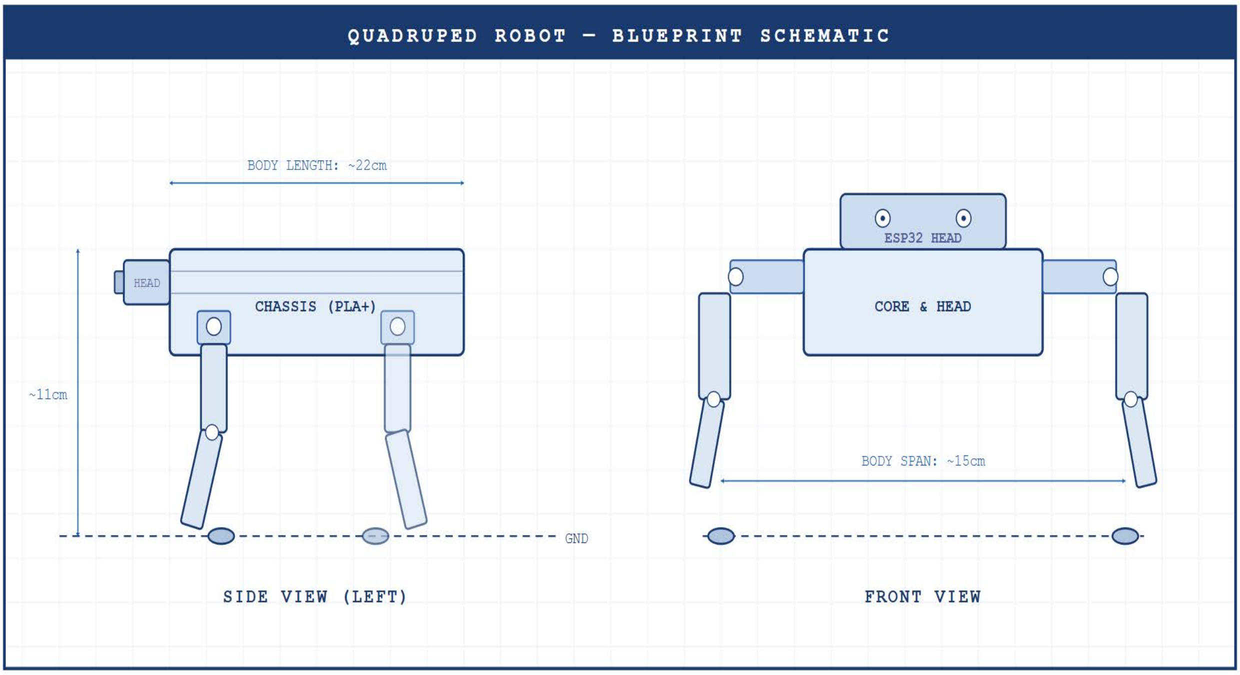


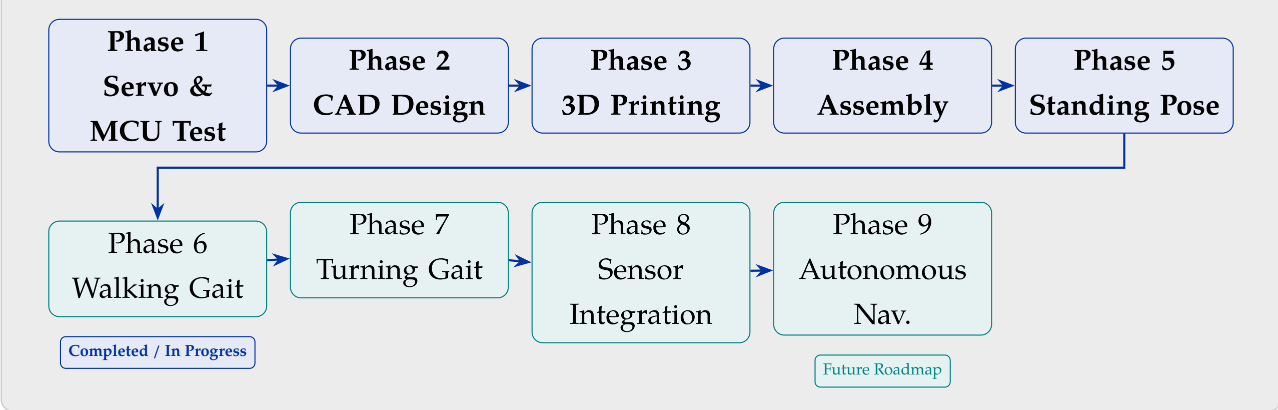
Figure 2: A Quadruped Robot (reference)

4. Journey from Concept to Assembled Robot

4.1. Overview of Development Phases

The project follows a structured nine-phase roadmap, progressing from individual component verification to autonomous navigation. Each phase builds on the previous one. Phases 1–5 are covered in detail here; Phases 6–9 form the future roadmap.

Development Roadmap — Phase Flowchart



4.2. Phase 1 — Component Verification

Before mechanical construction, each component was tested in isolation. The ESP32 was verified with an LED blink sketch. The PCA9685 was tested using an I2C scanner, confirming address `0x40`. All eight SG90 servos were tested directly on ESP32 GPIO18. A critical issue was encountered: servos were not moving when controlled through the PCA9685. Diagnosis revealed the **V+ power rail measured only 0.10 V** — the servo rail had no external 5 V supply. This was resolved by wiring 5 V directly to the V+ terminal.

4.3. Phase 2 — CAD Design

With dimensions finalised (see Section 6), the team modelled each component in FreeCAD using the Part Design workbench. Bodies were created for: Core Plate, Head Box, Upper Leg, Lower Leg, Servo Mount, and Assembly. The Core Plate was modelled first as it defines all hip servo mount positions.

4.4. Phase 3 — 3D Printing

All components were exported as STL files and sliced with standardised settings: 0.2 mm layer height, 40% infill, 4 perimeter walls, 0.4 mm nozzle. Core Plate and Head Box were printed in PLA+ for rigidity; leg segments in PETG for flexibility and impact resistance.

4.5. Phase 4 & 5 — Assembly and Standing Pose

Servo horns were attached to each SG90 and press-fitted into printed mounts. All eight servos were connected to the PCA9685 per the channel mapping (CH0–CH8). The initial standing pose was achieved by calibrating each servo's neutral angle to position the legs vertically, distributing the robot's weight evenly.

5. Electronics Architecture

5.1. Component Overview

The electronics stack has four layers: the **ESP32** as central controller, the **PCA9685** as servo driver (I2C), **eight SG90 servos** as actuators, and an **external 5 V supply** for the servo rail.

Component	Qty	Role
ESP32 Dev Board	1	Main controller (I2C master)
PCA9685 16-ch PWM Driver	1	Servo PWM generation (I2C slave)
SG90 Servo Motor	8	Actuators (legs + head)
5V External Power Supply	1	Servo rail power

5.2. ESP32 to PCA9685 I2C Connections

ESP32 Pin	PCA9685 Pin
3.3V	VCC
GND	GND
GPIO21	SDA
GPIO22	SCL

The PCA9685 uses 3.3 V logic but its servo rail (V+) must receive a **separate external 5 V supply**. Power for VCC and V+ are electrically isolated; the only shared connection is GND.

5.3. Servo Channel Mapping

Channel	Servo Function	Type
CH0	Front Left Hip	Hip
CH1	Front Left Knee	Knee
CH2	Front Right Hip	Hip
CH3	Front Right Knee	Knee
CH4	Rear Left Hip	Hip
CH5	Rear Left Knee	Knee
CH6	Rear Right Hip	Hip
CH7	Rear Right Knee	Knee
CH8	Head Rotation	Head

5.4. Servo Power Rail — Critical Discovery

During Phase 1 testing, an important real-world debugging experience was encountered:

Critical Finding: PCA9685 V+ Power Rail Issue

Symptom: All 8 servos unresponsive when controlled via PCA9685.

I2C Diagnosis: Scanner confirmed PCA9685 at address `0x40` — ESP32, I2C bus, and chip all functioning correctly.

Root Cause: V+ rail measured **0.10 V** (expected 4.8–5.2 V). Servo power rail not connected to external 5 V supply.

Resolution: Connect 5 V directly to PCA9685 V+ terminal. Servos responded immediately.

Lesson: Always verify servo rail voltage before suspecting firmware or I2C issues.

5.5. Direct Servo Testing Connections

Servo Wire	ESP32 Connection
Signal (Orange/Yellow)	GPIO18
VCC (Red)	VIN (5V)
GND (Brown/Black)	GND

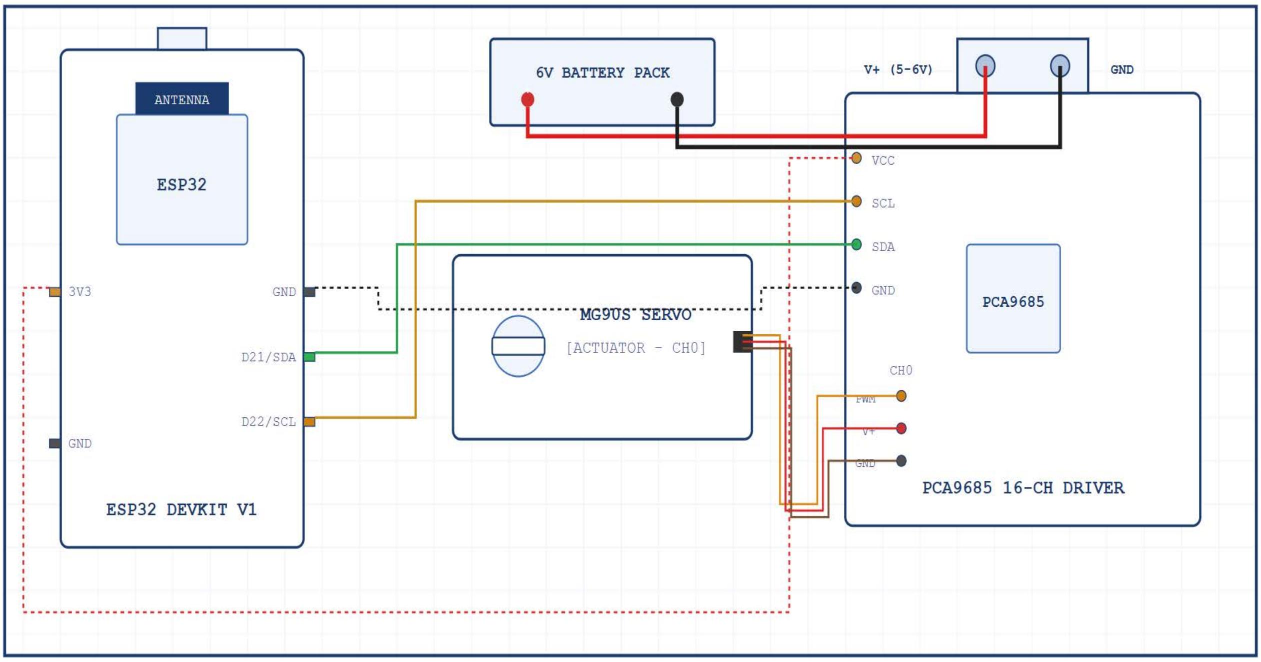


Figure 3: ESP32 and PCA9685 wiring setup

6. Mechanical Architecture & CAD Design

6.1. Structural Overview

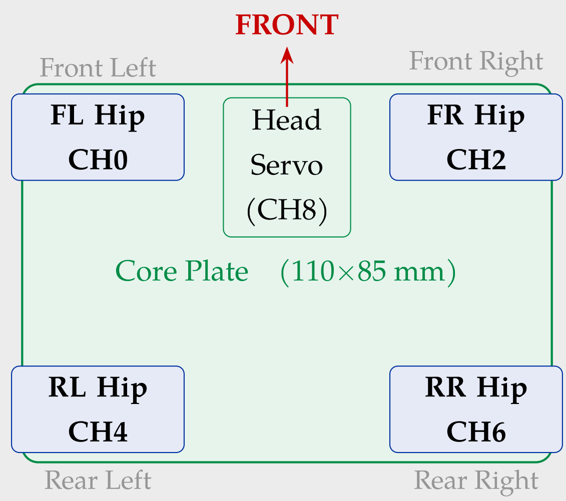
The quadruped's structure consists of two assemblies: the **Head Box** (housing all electronics) and the **Core Plate** (central chassis mounting all four hip servos). Each leg has two links — an **Upper Leg** driven by the hip servo and a **Lower Leg** driven by the knee servo.

6.2. Component Dimensions

Component	Dimensions (mm)	Notes
Core Plate	$110 \times 85 \times 4$	4 hip mounts + 1 head servo mount
Head Box	$85 \times 65 \times 45$	Wall 2.5 mm; ESP32, PCA9685, switch
Upper Leg	$45 \times 10 \times 5$	$\times 4$ total
Lower Leg	$55 \times 10 \times 5$	$\times 4$ total
Servo Mount	Custom	Press-fit for SG90 body

6.3. Servo Placement — Top View

Robot Top-View — Servo Layout Diagram



6.4. FreeCAD Design Workflow

1. **Core Plate** — Base dimensions with servo mounting pockets.
2. **Head Servo Mount** — Centred on Core Plate; controls head rotation.
3. **Hip Servo Mounts** — Four symmetric pockets at FL, FR, RL, RR.

4. **Upper Leg** — 45 mm link with servo horn connection hole.
5. **Lower Leg** — 55 mm link with foot pad boss at terminal end.
6. **Head Box** — Enclosure around ESP32 and PCA9685 PCB footprints.
7. **Assembly** — All bodies assembled with correct spacing.

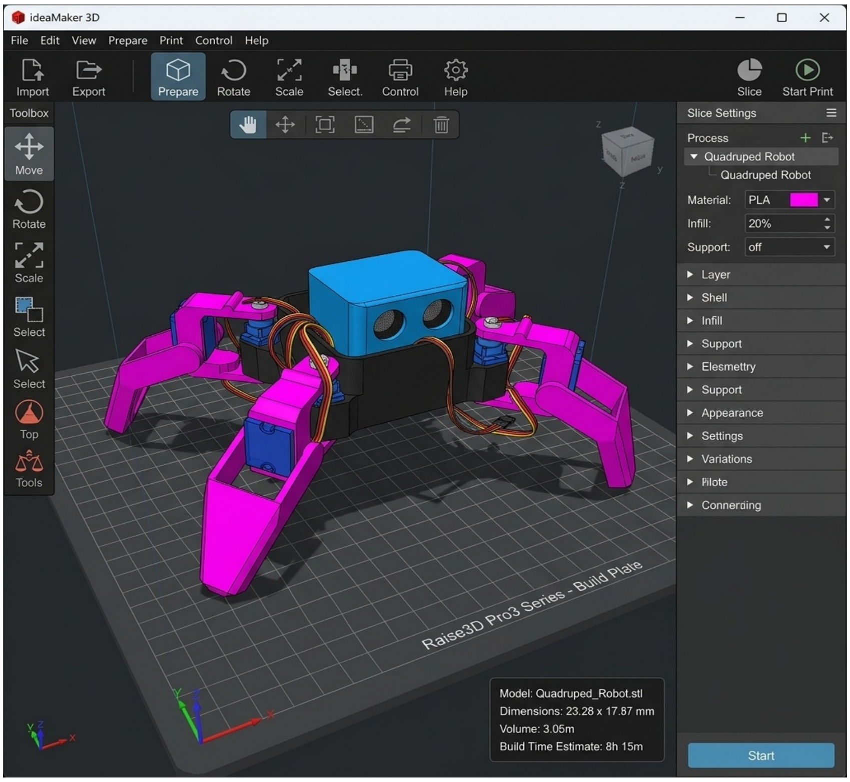


Figure 4: Full Assembly in Ideamaker

7. Products Made (Printed Parts)

7.1. Printed Part Summary

Part	Material	Dimensions	Qty
Core Plate	PLA+	$110 \times 85 \times 4$ mm	1
Head Box	PLA+	$85 \times 65 \times 45$ mm (wall 2.5)	1
Upper Leg	PETG	$45 \times 10 \times 5$ mm	4
Lower Leg	PETG	$55 \times 10 \times 5$ mm	4
Servo Mount	PLA+	Custom (SG90 press-fit)	9

7.2. Core Plate

The Core Plate is the central structural member — a flat 110×85 mm plate with four symmetric hip servo pockets and one central head servo mount. The 4 mm thickness provides bending rigidity for the servo loads.

7.3. Head Box

The Head Box (85×65×45 mm, wall 2.5 mm) houses the ESP32, PCA9685, and power switch, with provision for future sensor mounts.

7.4. Leg Components

Upper Leg (45 mm) and Lower Leg (55 mm) are PETG links, 10 mm wide and 5 mm thick, with servo horn attachment holes at the joint ends.

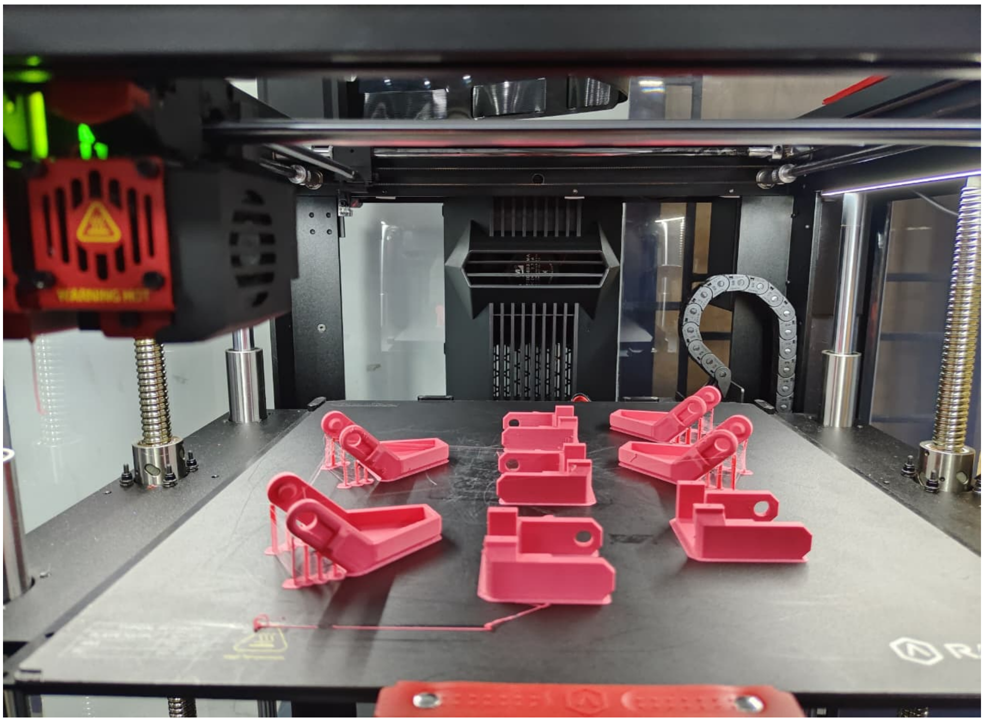


Figure 5: 3D Printed Leg Components (Material: PLA+)

8. Arduino Programming

8.1. Development Environment

All firmware was written using **Arduino IDE** with the ESP32 board support package. Libraries used:

- `ESP32Servo` — direct servo control via GPIO PWM.

- `Wire` — I2C communication for the PCA9685.
- `Adafruit_PWMServoDriver` — high-level PCA9685 control.

8.2. Servo Test Code

Each SG90 servo was tested individually on GPIO18 before PCA9685 integration:

```cpp
#include <ESP32Servo.h>

Servo myServo;

void setup() {
  myServo.attach(18);    // Signal on GPIO18
}

void loop() {
  myServo.write(0);      // 0 degrees
  delay(1000);
  myServo.write(90);     // 90 degrees (neutral)
  delay(1000);
  myServo.write(180);    // 180 degrees
  delay(1000);
}
```

Listing 1: Servo Test Code (ESP32 direct GPIO18)

8.3. PCA9685 I2C Scanner

Before running servo control through the PCA9685, the I2C bus was scanned to confirm the PCA9685 address:

```cpp
#include <Wire.h>

void setup() {
  Wire.begin(21, 22);    // SDA=GPIO21, SCL=GPIO22
  Serial.begin(115200);
  Serial.println("Scanning I2C bus...");
  for (byte addr = 1; addr < 127; addr++) {
    Wire.beginTransmission(addr);
    if (Wire.endTransmission() == 0) {
      Serial.print("I2C device found at 0x");
      if (addr < 16) Serial.print("0");
      Serial.println(addr, HEX);
    }
```

```
  }
  Serial.println("Scan␣complete.");
}

void loop() {}
```

Listing 2: I2C Scanner — PCA9685 Address Verification

Expected serial monitor output:

```
Scanning I2C bus...
I2C device found at 0x40
Scan complete.
```

Listing 3: I2C Scan Output — Serial Monitor

Address `0x40` is the factory-default PCA9685 I2C address (all A0–A5 pins low), confirming SDA/SCL wiring and chip operation.

8.4. PCA9685 Servo Control

Once the V+ issue was resolved (Section 5), all eight servos were driven through the PCA9685:

```
#include <Wire.h>
#include <Adafruit_PWMServoDriver.h>

Adafruit_PWMServoDriver pwm = Adafruit_PWMServoDriver(0x40);

#define SERVOMIN  150   // ~0 degrees @ 50Hz
#define SERVOMAX  600   // ~180 degrees @ 50Hz
#define SERVO_FREQ 50

uint16_t angleToPulse(int angle) {
  return map(angle, 0, 180, SERVOMIN, SERVOMAX);
}

void setup() {
  Wire.begin(21, 22);
  pwm.begin();
  pwm.setOscillatorFrequency(27000000);
  pwm.setPWMFreq(SERVO_FREQ);
  delay(10);
}

void loop() {
```

```
  for (int ch = 0; ch < 9; ch++)
    pwm.setPWM(ch, 0, angleToPulse(90));   // all neutral
  delay(2000);
  for (int ch = 0; ch < 9; ch++)
    pwm.setPWM(ch, 0, angleToPulse(45));   // all 45 deg
  delay(2000);
}
```

Listing 4: PCA9685 All-Channel Servo Sweep

9. Results & Testing

9.1. I2C Communication Verification

The PCA9685 I2C address was confirmed as `0x40` using the scanner (Listing 2), validating the full chain: ESP32 → GPIO21/22 → SDA/SCL → PCA9685.

9.2. Power Rail Diagnosis and Resolution

As documented in Section 5, the V+ rail measured 0.10 V initially. After connecting an external 5 V supply, it measured 4.92–5.05 V. All servos responded immediately.

9.3. Individual Servo Testing Results

All eight SG90 servos were tested using the direct sweep code (Listing 1):

Servo	Function	0°	180°
1	Front Left Hip (CH0)	Pass	Pass
2	Front Left Knee (CH1)	Pass	Pass
3	Front Right Hip (CH2)	Pass	Pass
4	Front Right Knee (CH3)	Pass	Pass
5	Rear Left Hip (CH4)	Pass	Pass
6	Rear Left Knee (CH5)	Pass	Pass
7	Rear Right Hip (CH6)	Pass	Pass
8	Rear Right Knee (CH7)	Pass	Pass
9	Head Rotation (CH8)	Pass	Pass

All 8 servos completed the full range 0°–90°–180° without issues.

9.4. Development Phase Status

Phase	Description	Status
1	Component verification	Complete
2	CAD design in FreeCAD	Complete
3	3D printing all parts	Complete
4	Servo calibration	Complete
5	Standing pose	Complete
6	Walking gait	In Progress
7	Turning gait	Planned
8	Sensor integration	Planned
9	Autonomous navigation	Future

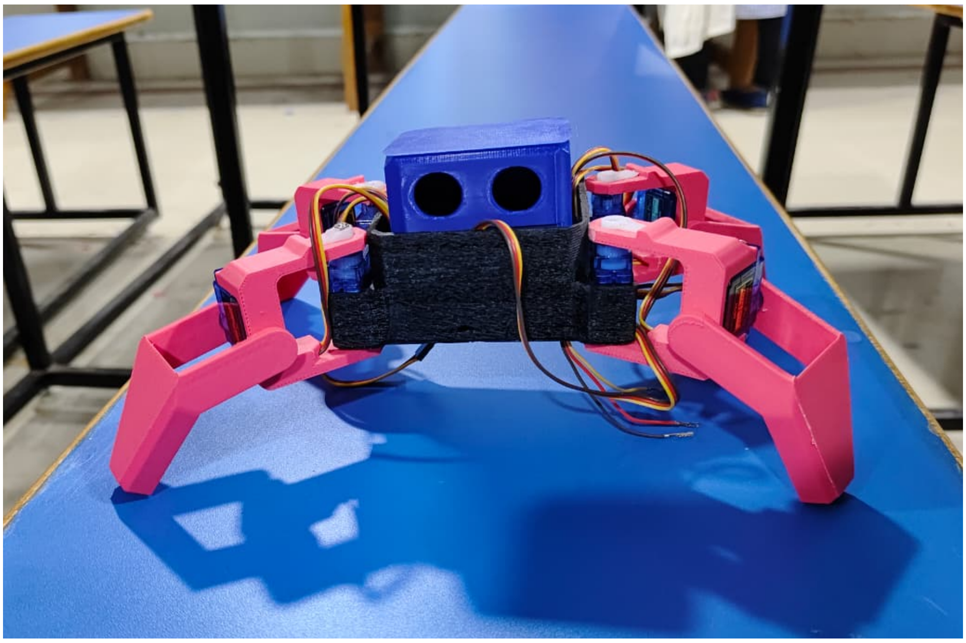


Figure 6: Partially Assembled Quadruped Robot

10. Conclusion

This project has been a comprehensive experience in legged robot development — from component verification through mechanical design, additive manufacturing, and embedded programming. Group-E has successfully built the hardware and software stack for an 8-DOF quadruped robot across three engineering disciplines.

From an **electronics perspective**, the project reinforced systematic debugging skills. The PCA9685 V+ rail issue — diagnosed through I2C scanning and multimeter probing — exemplifies real-world troubleshooting. All eight servos and the confirmed I2C address `0x40` provide a solid foundation.

From a **mechanical design perspective**, parametric CAD in FreeCAD taught servo mount constraints, press-fit tolerancing, and STL export for manufacturing.

From a **3D printing perspective**, material selection (PLA+ vs PETG), slicer optimisation, and print quality assessment gave hands-on applied knowledge building on the theory covered in this report.

Looking ahead: walking gait (Phase 6), turning (Phase 7), sensor integration (Phase 8), and autonomous navigation (Phase 9). The ESP32's Wi-Fi also enables remote teleoperation. Group-E is proud to have contributed a fully functional quadruped robot platform to the WTL & MeitY programme at CGEC.

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